

Evaluation of the CPF Badge Game-Theoretic Model

Summary

The model can explain each of the four findings individually, and it can jointly explain $\{1, 2, 3\}$ or $\{1, 2, 4\}$. However, it **cannot explain all four simultaneously**. The obstruction is sharp: once Finding 3 forces the platform to choose the selective-badging regime (situation 2), Findings 3 and 4 impose *contradictory* requirements on the same parameter, $\alpha_{A0} - \alpha_{B0}$. Below I derive the equilibrium, map each finding to it, prove the impossibility, and then propose modifications that remove it.

1 How the model works

The model is a three-stage game (the platform sets the badge threshold; the two sellers set prices in differentiated Bertrand competition; consumers choose quantities) built on Singh–Vives (1984) linear demand. Solving the consumer problem (FOCs on quantities) gives linear demand, and solving each seller’s problem (FOC on price) gives the reaction function $p_j = (X_j + z p_{-j})/2$, where X_j is a demand intercept. Imposing symmetry yields the equilibrium price

$$p_A^{(1)} = \frac{(2 - z^2) \alpha_{A0} - z \alpha_{B0}}{4 - z^2}.$$

The three regimes (situation 1 = no badge, 2 = only A badged, 3 = both badged) differ *only* in the intercept X_j : badging adds a green-consumer term $k\rho_0$ to the intercept of the badged product. Every equilibrium is therefore the same solution with a shifted intercept, which is why all three Lemmas share the denominator $4 - z^2$.

2 Mapping the four findings to the equilibrium

Finding 2 (price increases) — unconditional. Directly from the shifted intercept,

$$p_A^{(2)} - p_A^{(1)} = \frac{2k\rho_0}{4 - z^2} > 0, \quad p_B^{(2)} - p_B^{(1)} = \frac{zk\rho_0}{4 - z^2} > 0.$$

A raises its price because the badge attracts green consumers; B follows upward through strategic complementarity. No condition is needed.

Finding 1 (demand increases) — conditional. $D_A^{(2)} > D_A^{(1)}$ requires (eq. 25)

$$(2 - z^2) \alpha_{A0} > z \alpha_{B0} + (1 - z^2) \rho_0,$$

i.e. the green-consumer inflow must outweigh the non-green consumers driven away by the price premium. B ’s demand rises unconditionally (it gains a relative price advantage once A raises its price).

Finding 3 (platform prefers selective badging = situation 2 optimal) — knife-edge.

This is Proposition 3. The platform maximizes total revenue $p_A D_A + p_B D_B$. Selective badging is attractive because asymmetric badging *differentiates* the two products and softens price competition (in situation 3 both raise prices symmetrically, so the rivalry stays fierce). The proof requires that *every* component— p_A, p_B, D_A, D_B —be highest in situation 2. Combining these inequalities pins α_{A0} into a narrow band slightly **below** α_{B0} (for $z = 0.8$: $\alpha_{B0} - 0.44 < \alpha_{A0} < \alpha_{B0} - 0.10$), together with $0.67 < z < 0.89$. Intuitively, since the badged product A already receives the badge’s advantage for free, its intrinsic quality must be slightly *weaker* so that the two sellers remain evenly matched.

Finding 4 (HHI decreases). With two firms, $\text{HHI} = MS_A^2 + (1 - MS_A)^2$, minimized at $MS_A = 0.5$; hence HHI falls **iff** $|MS_A - \frac{1}{2}| = \frac{|D_A - D_B|}{2(D_A + D_B)}$ shrinks, i.e. shares move toward 50/50. There are two routes:

- *Situation 3 (both badged)*: the symmetric badge shift enlarges total demand $D_A + D_B$ and dilutes the pre-existing share asymmetry, so shares converge — HHI falls almost unconditionally (the paper’s Proposition 5).
- *Situation 2 (only A badged)*: asymmetric. Proposition 4 requires $\alpha_{A0} \gg \alpha_{B0}$ (numerically 64 vs 8). The mechanism is counterintuitive: when A is overwhelmingly dominant it is already near-monopoly (share far from 0.5); the badge forces A to raise its price, which pushes non-green consumers toward the weaker B , *pulling the inflated share back toward 0.5*.

3 Why the model cannot satisfy all four simultaneously

Finding 3 asserts that the regime the platform actually selects is **situation 2**. The market-concentration claim (Finding 4) must therefore be evaluated *in situation 2*—one cannot let the platform play situation 2 while reporting the HHI of situation 3. In situation 2:

- Finding 3 (Prop. 3) requires α_{A0} to be **slightly smaller** than α_{B0} (so that asymmetric badging is the platform’s optimum);
- Finding 4 (Prop. 4) requires α_{A0} to be **much larger** than α_{B0} (so that A ’s badge-induced price hike pulls its near-monopoly share back toward 0.5).

These are requirements on the *same* quantity $\alpha_{A0} - \alpha_{B0}$: one needs it slightly negative, the other needs it large and positive. The two regions are disjoint. The contrast is stark in the paper’s own numerical examples—Prop. 3 uses $\alpha_{A0} \in (1.56, 1.90) < \alpha_{B0} = 2$, while Prop. 4 uses $\alpha_{A0} = 64 \gg \alpha_{B0} = 8$. **No parameter configuration makes situation 2 both the platform’s optimum and HHI-reducing**, so the four findings cannot hold at once. The deeper reason: the only robust HHI-reducing path (situation 3, full badging) is precisely the regime that Finding 3 and the empirical “ $\approx 50\%$ optimal threshold” rule out.

4 Supporting criticisms

1. **The Prop. 4 example leaves the model’s valid region.** With $\alpha_{A0} = 64$, $\alpha_{B0} = 8$, $z = 0.5$, the no-badge equilibrium price for B , obtained by substituting this example into the Lemma 1 price formula, is $p_B^{(1)} = -4.8 < 0$, violating the paper’s own non-negativity constraint (eq. 17). In that example B is effectively excluded and the market is a monopoly (HHI = 1, maximal concentration)—the opposite of the claimed result.

2. **HHI is too low-dimensional in a duopoly.** “HHI falls” here means only “two incumbents move toward 50/50,” with no entry, exit, or small-seller channel—yet the empirical Finding 4 is told as “small brands catch up,” and section 6.3 even *drops* small brands when computing HHI while Table 6 shows the *largest* brands have the strongest demand response (-0.164 vs -0.089 for small brands), an internal tension.
3. **The optimality proof is needlessly strong.** Requiring every price and quantity to be highest in situation 2 is a *sufficient but far from necessary* condition for total revenue to be highest (revenue can be maximal even if one component is lower). This artificially shrinks the feasible region and is the main reason Proposition 3 is a knife-edge; the paper concedes this in footnote 4.
4. **The badge carries zero information.** Sustainability f_j is exogenous and badging is costless, so “selective badging is optimal” is purely a price-competition by-product rather than a result of certification economics (scarcity $\rightarrow$ information value), which is what actually drives platforms to set thresholds.
5. **Minor error.** Equation (41) states $\text{HHI} \geq 0.25$, but for $n = 2$ the lower bound is $0.5^2 + 0.5^2 = 0.5$; 0.25 is the $n \rightarrow \infty$ bound.

5 Modifications that allow all four to hold simultaneously

The goal is to dismantle the disjoint-condition obstruction. Each modification targets one root cause.

Modification A — Relax the assumption that green consumers buy *only* badged products. *Problem:* the all-or-nothing rule means an unbadged B captures zero green demand, creating the extreme asymmetry in situation 2 that forces $\alpha_{A0} \gg \alpha_{B0}$ for HHI to fall. *Fix:* let ρ enter utility continuously—green consumers still buy the unbadged B , merely preferring the badged A . *Why it works:* B is no longer dropped off a cliff, so the badge’s effect on the share gap becomes a smooth slope rather than a jump. HHI can then fall even when $\alpha_{A0} \lesssim \alpha_{B0}$ —exactly the region Finding 3 requires. This directly eliminates the sign conflict between Findings 3 and 4.

Modification B — Move from duopoly to many sellers (or add a competitive fringe of small sellers). *Problem:* with only A and B , “HHI falls” can only mean “two incumbents converge to 50/50”; the empirical mechanism (“small brands become more competitive”) has no place in the model. *Fix:* introduce small sellers and model selective badging as “the threshold is set so that some *smaller* sellers cross it and receive the badge.” *Why it works:* the badge then directly raises small sellers’ shares and lowers HHI, giving Finding 4 a mechanism that matches the data *and* is naturally compatible with selective badging. With more than two sellers, “selectivity” and “falling concentration” are no longer forced through the single 50/50 channel.

Modification C — Give the badge informational value or an issuance cost, so selectivity is endogenously optimal. *Problem:* the badge is costless and uninformative, so Finding 3 rests on Prop. 3’s fragile knife-edge and is merely a price-competition artifact. *Fix:* let consumers not observe true sustainability f_j , so the badge is a certification signal, and let full badging *dilute* its credibility (the effective premium $\rho_{\text{eff}}(I_0)$ decreases in the badged share). *Why it works:* the

platform now trades off “badge more $\rightarrow$ collect more commission” against “badge more $\rightarrow$ dilute the badge.” This generates an *interior* optimal threshold $I^* \in (0, 1)$ that replaces the knife-edge, turning Finding 3 into a robust certification-economics result that no longer depends on the ordering of $\alpha_{A0} - \alpha_{B0}$, and hence no longer competes with Finding 4 for parameters.

Why the three together work. A removes the forced asymmetry (so Finding 4 no longer needs $\alpha_{A0} \gg \alpha_{B0}$ and stops clashing with Finding 3’s “slightly weaker”); B gives Finding 4 a data-consistent mechanism compatible with selectivity; C upgrades Finding 3 from a fragile coincidence to a robust interior optimum. Together they open an *open set* of parameters—not a single point—in which selective badging (F3) + higher prices (F2) + higher demand (F1) + lower concentration (F4) all hold at once. **The minimal viable fix is A + B**, which dissolves the core contradiction; **adding C** is what makes Finding 3 itself robust.